CS 544: Optimization for ML - MP5

Xiaoyu Zhang (zhang401), Mayank Shrivastava (mayanks4),

The-Anh Vu-Le (vltanh), Samuel Schapiro (sjs17)

May 13th, 2024

1 Introduction

In the final project, we studied two open problems related to Sharpness-Aware Minimization (SAM), an optimizer that seeks flat minima for improved generalization. SAM has been considerably successful in practice, reaching state-of-the-art accuracies on SVHN, CIFAR-10, and more. However, since SAM is a recent development (published initially to arXiv in late 2020), many open questions exist. For our final project, we studied the following two questions:

1. In the original SAM formulation, the ρ hyperparameter (introduced in Section 3.2) is held constant for every time step. Instead, *should ρ be scheduled throughout training?*
2. *Does SAM dampen the group-wise loss oscillations due to the "outliers with opposing signals phenomenon"* (introduced in Section 2.3), whereby a network will learn statistically weak features that are unlikely to generalize and induce opposing gradients on distinct groups of input data points?

Outline The rest of this report is structured as follows: in Section 2, we provide background on flat minima, the edge of stability, and opposing signals; in Section 3, we introduce the SAM optimizer; in Section 4, we discuss our experiments with ρ scheduling; in Section 5, we discuss our experiments with the opposing signals phenomenon; and then in Section 6, we summarize our main insights from this project.

2 Related works

2.1 Flat Minima

The study of flat minima in relation to generalization dates back to at least [Hochreiter and Schmidhuber, 1997]. Since then, many works have continued to explore algorithms that lead to flatter solutions [Foret et al., 2021, Wortsman et al., 2022, Chaudhari et al., 2017], how various hyperparameter choices affect sharpness [Keskar et al., 2017] and have sought richer theoretical foundations for SAM [Bartlett et al., 2023, Behdin and Mazumder, 2023, Wen et al., 2023, Chen et al., 2023]. Recently, [Wen et al., 2023] have found that the flatness of the final solution does not sufficiently capture the generalization benefit from SAM alone, suggesting a more thorough study into the qualitative properties of features selected by SAM. Along these lines, [Chen et al., 2023] have shown that SAM has an implicit denoising mechanism which leads to benign overfitting in settings where SGD would harmfully overfit, and [Andriushchenko et al., 2023] have found that SAM prunes ReLU units and leads to lower-rank features.

2.2 Progressive Sharpening and the Edge-of-Stability

In [Cohen et al., 2022], the authors characterize a behavior of gradient descent on neural networks (NNs) known as progressive sharpening, whereby the *sharpness* (maximum eigenvalue of the loss Hessian) increases continually until reaching $2/\eta$, where η is the learning rate. Once sharpness crosses this threshold, optimization enters the *edge of stability*, a point where the sharpness will hover around $2/\eta$ while the training loss continues to decrease, albeit non-monotonically. The authors empirically observe this effect on multiple architectures.

2.3 Opposing Signals

In [Rosenfeld and Risteski, 2023], the authors hypothesize that *opposing signals* can cause the edge of stability. Spurious features are learned because they are correlated with the task labels. For example, photos of planes are likely to contain a blue background, but classifying images with large regions of blue as planes is not always correct, e.g., ships in the ocean. During optimization, decreasing loss on one of the opposing groups causes the loss of the others to rise, thus the name opposing signal.

3 Sharpness-Aware Minimization

3.1 Preliminaries

We consider a parametric model family $\Theta \subset \mathbb{R}^k$ with each model defined with respect to a specific choice of parameter $\theta \in \Theta$. The classifier induced from a model θ is denoted $f_\theta : \mathcal{X} \to \mathcal{Y}$.

We consider the pointwise loss function $\ell : \mathcal{Y} \times \mathcal{X} \times \Theta \to \mathbb{R}^+$. We define the population and empirical risk with respect to parameters $\theta \in \Theta$ as $\mathcal{L}(\theta) := \mathbb{E}_{\mathbf{x},y\sim\mathcal{D}}[\ell(y, \mathbf{x}, \theta)]$ and $\hat{\mathcal{L}}(\theta) := \frac{1}{n}\sum_{i=1}^{n} \ell(y_i, \mathbf{x}_i, \theta)$, respectively, for some distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$.

3.2 Sharpness-Aware Minimization

Introduced by [Foret et al., 2021], sharpness-aware minimization (SAM) proposes minimizing a version of the robust empirical loss rather than the standard loss. This is motivated by the following PAC-Bayesian bound, whose proof can be found in the appendix of Foret et al. [2021].

Theorem 3.1 (SAM Motivation)**.** For any $\rho > 0$ and model $\theta \in \Theta \subset \mathbb{R}^k$, if $\mathcal{L}(\theta) \leq \mathbb{E}_{\epsilon_i\sim\mathcal{N}(0,\rho)}[\mathcal{L}(\theta + \epsilon)]$, then w.p. $\geq 1 - \delta$,

$$\mathcal{L}(\theta) \leq \max_{\|\beta\|_2 \leq \rho} \hat{\mathcal{L}}(\theta + \beta) + \mathcal{O}\left(\sqrt{\frac{k \ln(\|\theta\|_2^2/\rho^2) + \ln(n/\delta)}{n}}\right)$$

Thus, SAM's objective is to find the solution to the following optimization problem:

$$\min_{\theta\in\Theta} \max_{\beta:\|\beta\|_2 \leq \rho} \mathcal{L}(\theta + \beta) \tag{1}$$

The inner maximization is difficult to solve exactly, so two approximations are used:

1. A first-order Taylor approximation of the loss is made so that the objective is of the form

$$\max_{\beta:\|\beta\|_2 \leq \rho} \mathcal{L}(\theta + \beta) \overset{(i)}{\approx} \max_{\beta:\|\beta\|_2 \leq \rho} \mathcal{L}(\theta) + \beta^\top \nabla_\theta \mathcal{L}(\theta) = \max_{\beta:\|\beta\|_2 \leq \rho} \beta^\top \nabla_\theta \mathcal{L}(\theta) \tag{2}$$

   The solution to Equation (2) is the classical dual norm solution

$$\beta^\star(\theta) := \arg\max_{\beta:\|\beta\|_2 \leq \rho} \beta^\top \nabla_\theta \mathcal{L}(\theta) = \rho \frac{\nabla_\theta \mathcal{L}(\theta)}{\|\nabla_\theta \mathcal{L}(\theta)\|} \tag{3}$$

2. Then, the computation of the SAM gradient $\nabla_\theta \mathcal{L}\left(\theta + \beta^\star(\theta)\right)$ drops the second-order term relating to the derivative of the perturbation with respect to the weights. Therefore, by chain rule, the SAM gradient is the following:

$$\nabla_\theta \mathcal{L}\left(\theta + \beta^\star(\theta)\right) = \frac{\mathrm{d}\theta}{\mathrm{d}\theta} \nabla_\theta \mathcal{L}\left(\theta\right)|_{\theta+\beta^\star(\theta)} + \frac{\mathrm{d}\beta^\star(\theta)}{\mathrm{d}\theta} \nabla_\theta \mathcal{L}\left(\theta\right)|_{\theta+\beta^\star(\theta)} \tag{4}$$

$$\overset{(ii)}{\approx} \nabla_\theta \mathcal{L}\left(\theta\right)|_{\theta+\beta^\star(\theta)} \tag{5}$$

Therefore, the SAM update rule at each time step t is given by

$$\theta_{t+1} := \theta_t - \eta \nabla_\theta \mathcal{L}(\theta)|_{\theta+\rho\frac{\nabla_\theta \mathcal{L}(\theta)}{\|\nabla_\theta \mathcal{L}(\theta)\|}} \tag{6}$$

Intuitively, the practical implementation is a simple modification of standard gradient descent where instead of taking the gradient at θ, i.e., $\nabla_\theta \mathcal{L}(\theta)$, we first climb to $\theta+\rho\frac{\nabla_\theta \mathcal{L}(\theta)}{\|\nabla_\theta \mathcal{L}(\theta)\|}$, take the gradient $\nabla_\theta \mathcal{L}(\theta)|_{\theta+\rho\frac{\nabla_\theta \mathcal{L}(\theta)}{\|\nabla_\theta \mathcal{L}(\theta)\|}}$ at this point, and then use such gradient to perform the update on θ.

4 ρ Schedule Experiments

In Section 6, we derive the SAM update rule and show how it is a simple modification of gradient descent where gradients are instead taken at $\theta + \rho\frac{\nabla_\theta \mathcal{L}(\theta)}{\|\nabla_\theta \mathcal{L}(\theta)\|}$ to urge the model towards flatter regions of the loss landscape. The original SAM keeps ρ constant throughout training. In this section, we examine several schedules with varied values of ρ throughout training.

Training Specifications For each ρ schedule, we train a ResNet18 on CIFAR-10 dataset for 200 epochs. Since we used the same specifications as Foret et al. [2021], we used $\rho = 0.05$, which was chosen by their hyperparameter tuning. Let t denote the epoch index, starting from $t = 1$. Each ρ schedule is listed and motivated below:

1. **Constant** sets $\rho_t = 0.05$ for all $t \in [1, 200]$ and is the baseline we compare against.

2. **Oscillatory** sets

$$\rho_t = \begin{cases} 0.025 & \text{if } t \text{ is even} \\ 0.075 & \text{if } t \text{ is odd} \end{cases}$$

   and is motivated by the use of cyclical learning rate schedules, which are believed to help escape local minima during training. We hypothesize that oscillatory ρ schedules may have a similar effect and lead to better generalization.

3. **Decay Near End** sets

$$\rho_t = \begin{cases} 0.05 & t \leq 150 \\ 0.05 + (150 - t)/50 & t > 150 \end{cases}$$

   so that ρ gradually decays from 0.05 to 0 between $t = 151$ and $t = 200$. The motivation for this is that near the end of the training, we may wish to slowly return to SGD updates since SAM has converged to a flat region of the loss landscape, and our objective will be to go directly to the minimum of this loss landscape region.

4. **Increase Near End** sets

$$\rho_t = \begin{cases} 0.05 & t \leq 150 \\ 0.05 + (t - 150)/50 & t > 150 \end{cases}$$

   so that ρ gradually increases from 0.05 to 0.1 between $t = 151$ and $t = 200$. After discovering that the **decay near end** schedule was unsuccessful (in Table 1), we tried the exact opposite of this.

ρ **Schedule**	**Equation**	**Test Accuracy**
Constant	$\rho_t = 0.05$	**96.22**
Oscillatory	$\rho_t = \begin{cases} 0.025 & \text{if } t \text{ is even} \\ 0.075 & \text{if } t \text{ is odd} \end{cases}$	96.13
Decay Near End	$\rho_t = \begin{cases} 0.05 & t \leq 150 \\ 0.05 + (150 - t)/50 & t > 150 \end{cases}$	95.82
Increase Near End	$\rho_t = \begin{cases} 0.05 & t \leq 150 \\ 0.05 + (t - 150)/50 & t > 150 \end{cases}$	96.16

Table 1: A description and definition of each ρ schedule, accompanied by the highest test accuracy it obtains.

Results and Discussion As shown in Table 1, none of the ρ schedules we tried outperformed the original, constant $\rho = 0.05$. The **Decay Near End** schedule performed the worst, suggesting that *there is still significant benefit to using SAM even as the model is very close to convergence.* The **Oscillatory** schedule performed slightly worse than the constant setting, suggesting that these oscillations were not helpful in achieving better generalization. Finally, the **Increase Near End** schedule performed the closest to the original setting — but slightly worse — suggesting that gradually increasing ρ to twice its original value (0.1 versus 0.05) does not degrade performance by very much.

5 Opposing Signal Experiments

In this experiment, we want to replicate a toy example of fitting small multi-layer perceptrons (MLPs) to the Chebyshev polynomial (Cohen et al. [2022], Rosenfeld and Risteski [2023]) and compare the opposing signals phenomenon exhibited by stochastic gradient descent (SGD) and SAM. Since Long and Bartlett [2024] shows that SAM has a smaller edge of stability, we expect that SAM can reduce oscillation and lead to more stable training.

5.1 Problem Definition

The network is trained to model the following functions:

$$\text{Chebyshev}(k) : T_k(x) = \cos(kx), \forall x \in \mathbb{R} \qquad k = [3, 4, 5]$$

Each dataset contains 20 samples, $\{(x_i, y_i)\}_{i \leq 20}$, where x_i's are equally spaced between $[-1, 1]$. The datasets are shown in Figure 1. We define three groups in each dataset.

1. Group 1 contains points with $x_i < -0.8$,
2. Group 2 contains points with $-0.8 \leq x_i \leq -0.4$, and
3. Others contains the remaining samples.

Both Group 1 and Group 2 are samples whose x_i's are large negative values, but y_i's are distinctive. This partitioning is constructed to induce opposing signals. The large negativity of the input feature (x_i) is analogous to the blue background in both plane and ship images. During training, the loss of each group is tracked for the examination of the oscillation.

5.2 Network Architecture and Training Details

Considering the simplicity of the dataset, we follow previous works Cohen et al. [2022], Rosenfeld and Risteski [2023] and use small MLPs, a handful of linear layers followed by tanh activation. An example of

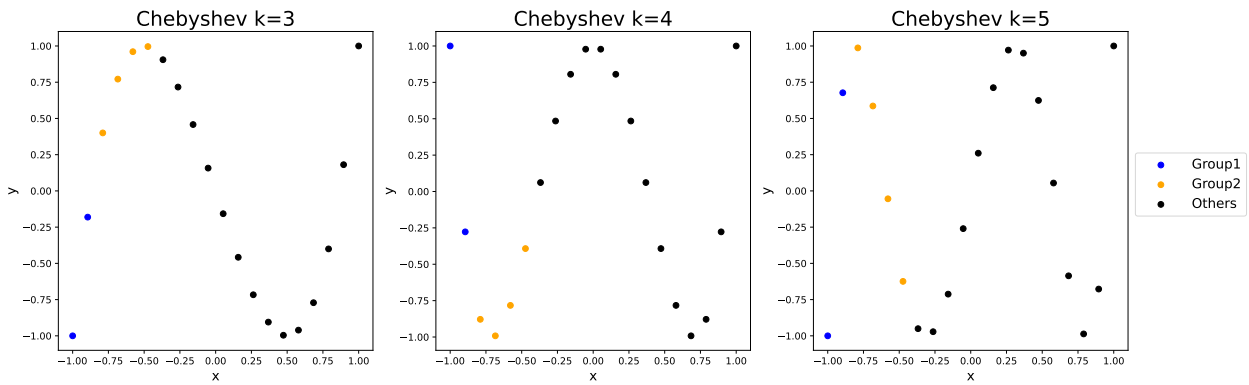


Figure 1: Chebyshev Polynomials with different k. Each dataset contains 20 samples.

a depth-4 MLP is:

```
Sequential(
    (0): Linear(in_features=1, out_features=200, bias=True)
    (1): Tanh()
    (2): Linear(in_features=200, out_features=200, bias=True)
    (3): Tanh()
    (4): Linear(in_features=200, out_features=200, bias=True)
    (5): Tanh()
    (6): Linear(in_features=200, out_features=200, bias=True)
    (7): Tanh()
    (8): Linear(in_features=200, out_features=1, bias=True)
)
```

We use MSE loss for training. The SGD optimizer is initialized with a learning rate of 0.01, momentum of 0.9, and weight decay 5×10^{-4}.

5.3 Results and Discussion

We first fix the dataset to be Chebyshev(3) and observe the loss curves of SGD and SAM across depth-$2, 3, 4$ MLPs. The results are shown in Figure 2, 3, and 4, respectively. The loss curves of SGD in these three figures show that instability occurs when network depth increases, but — on the other hand — for deeper networks, SAM avoids sudden spikes in losses and reduces loss more smoothly in later parts of training. Secondly, opposing signals happen in SGD at a much more prominent scale than that in SAM. Amplified opposing signal is best illustrated in Figure 4a: whenever loss of Group 1 increases, loss of Group 2 decreases. Lastly, the choice of ρ does not impact training; loss curves are similar for $\rho = 0.01, 0.5, 0.1$.

Figure 5 and 6 show behaviors of SGD and SAM for higher-order Chebyshev polynomials while keeping $\rho = 0.05$. For these two more complex datasets, the loss of SGD oscillates and spikes prominently, while SAM leads to a much smoother loss landscape. However, loss converges towards 0.0 more consistently in SGD than in SAM. In reality, a decrease in the training loss does not always translate to an increase in test performance. Further analysis and comparison of generalization between the two algorithms can be conducted in future works.

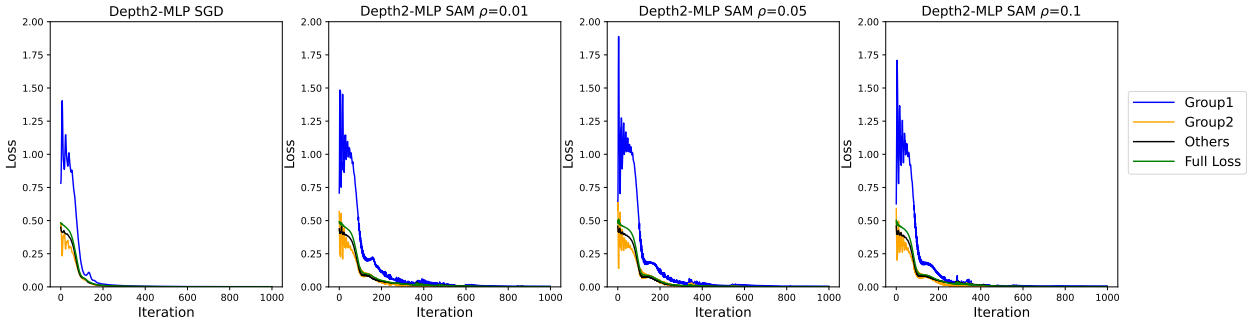


Figure 2: Loss curves for Chebyshev(3) using SGD and SAM. For this shallowest network with only 3 linear layers, SGD exhibits no significant oscillations.

6 Conclusion

In the final project, we studied the following two questions: (1) *Should ρ be scheduled throughout training for SAM?* and (2) *Does SAM dampen the group-wise loss oscillations due to the "outliers with opposing signals phenomenon"?*

To answer (1), we tried an oscillatory, decaying, and increasing ρ schedule. Although none of the proposed ρ schedules outperformed the constant ρ schedule on the chosen network architecture and dataset, this remains a promising research direction since there are various other schedules, architectures, and datasets to explore.

To answer (2), we train MLPs with different depths to points sampled on Chebyshev polynomials. We also track the training loss of two groups whose input x values are strongly negative but with different target outputs. Experiments show that SGD's loss curves exhibit significant instability during training; in contrast, SAM reduces loss more smoothly, although it sometimes takes longer to converge. Future direction includes tracking opposing signals in deeper networks with image inputs and comparing the generalization capability of the two optimization schemes.

References

M. Andriushchenko, D. Bahri, H. Mobahi, and N. Flammarion. Sharpness-aware minimization leads to low-rank features, 2023.

P. L. Bartlett, P. M. Long, and O. Bousquet. The dynamics of sharpness-aware minimization: Bouncing across ravines and drifting towards wide minima, 2023.

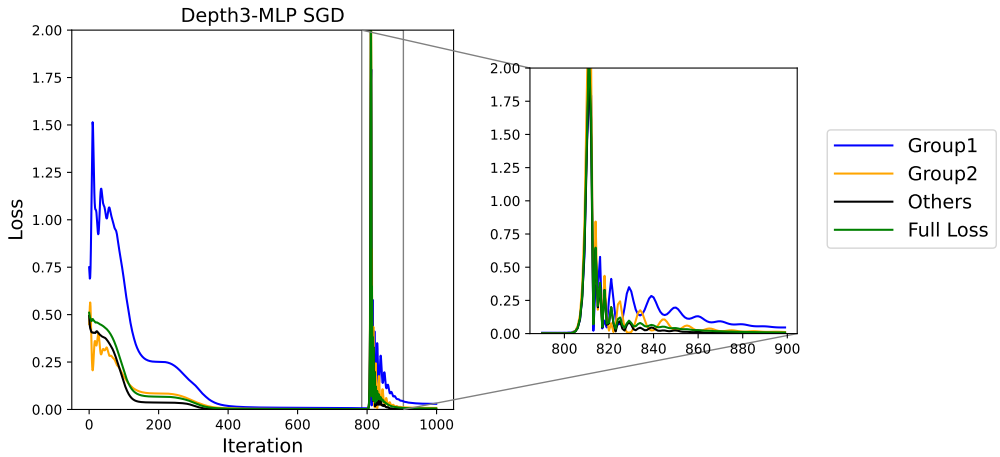
K. Behdin and R. Mazumder. On statistical properties of sharpness-aware minimization: Provable guarantees, 2023.

P. Chaudhari, A. Choromanska, S. Soatto, Y. LeCun, C. Baldassi, C. Borgs, J. Chayes, L. Sagun, and R. Zecchina. Entropy-sgd: Biasing gradient descent into wide valleys, 2017.

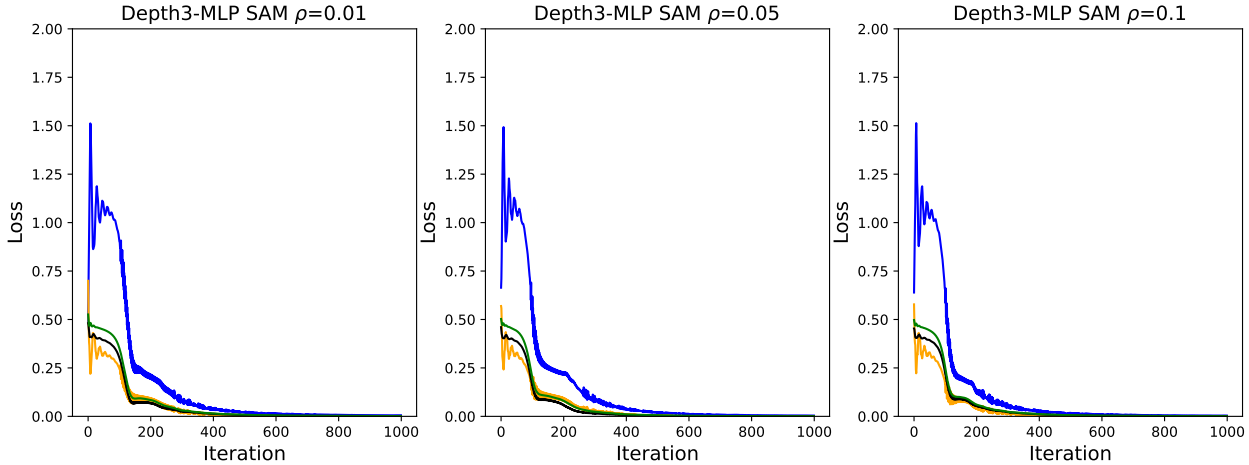
Z. Chen, J. Zhang, Y. Kou, X. Chen, C.-J. Hsieh, and Q. Gu. Why does sharpness-aware minimization generalize better than sgd?, 2023.

J. M. Cohen, S. Kaur, Y. Li, J. Z. Kolter, and A. Talwalkar. Gradient descent on neural networks typically occurs at the edge of stability, 2022.

P. Foret, A. Kleiner, H. Mobahi, and B. Neyshabur. Sharpness-aware minimization for efficiently improving generalization, 2021.

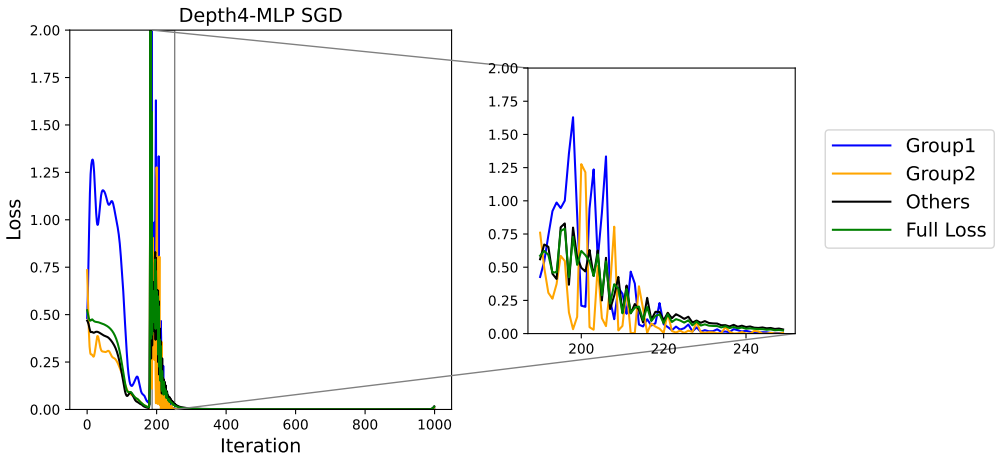


(a) Loss using SGD. Inset shows oscillation; comparing Group 1 and Group 2, we see the opposing signals phenomenon.

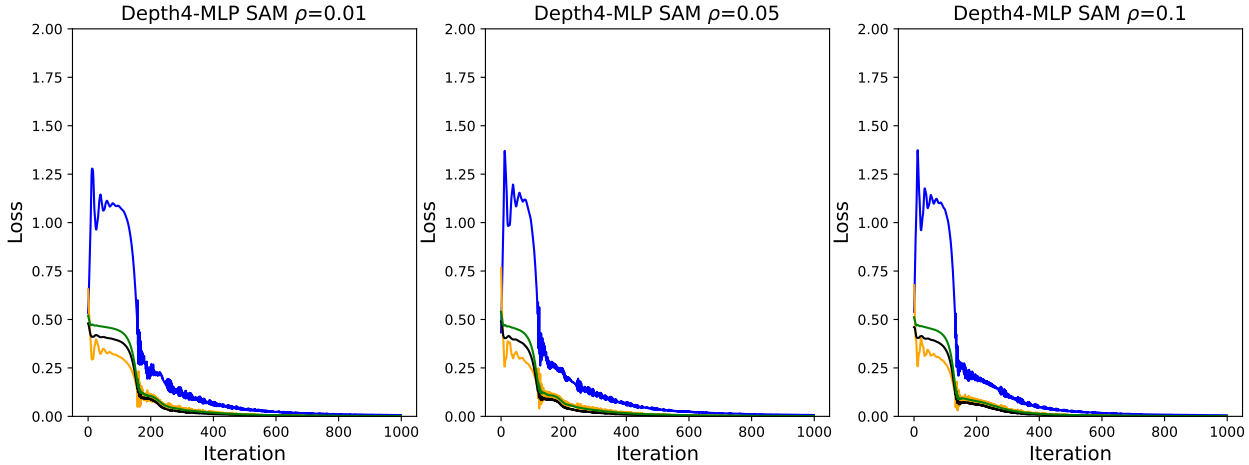


(b) Loss using SAM. Across different ρ values, SAM stabilizes the optimization process and reduces opposing signals. No divergence appears during training.

Figure 3: Results on optimizing a depth-3 MLP using SGD and SAM for Chebyshev(3).



(a) Loss using SGD. Inset shows oscillation; comparing Group 1 and Group 2, we see the opposing signals phenomenon.



(b) Loss using SAM. Across different ρ values, SAM stabilizes the optimization process and reduces opposing signals. No divergence appears during training.

Figure 4: Results on optimizing a depth-4 MLP using SGD and SAM for Chebyshev(3).

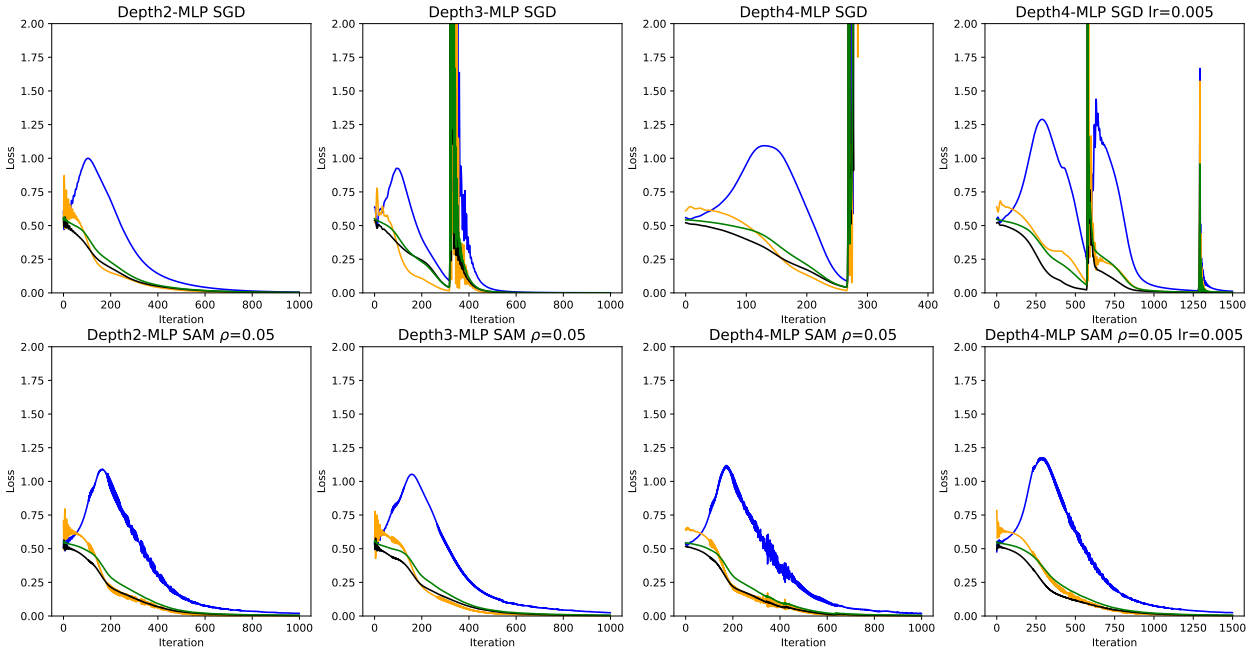


Figure 5: Loss curves for fitting $Chebyshev(4)$ using SGD and SAM. In the first three columns, learning rates are all 0.01. While SAM converges for MLPs of different depths, SGD diverges completely for depth-4 MLPs. If we lower the learning rate to 0.005, although SGD avoids divergence, the loss will still spike during the later part of training.

S. Hochreiter and J. Schmidhuber. Flat Minima. *Neural Computation*, 9(1):1–42, 01 1997. ISSN 0899-7667. doi: 10.1162/neco.1997.9.1.1. URL https://doi.org/10.1162/neco.1997.9.1.1.

N. S. Keskar, D. Mudigere, J. Nocedal, M. Smelyanskiy, and P. T. P. Tang. On large-batch training for deep learning: Generalization gap and sharp minima, 2017.

P. M. Long and P. L. Bartlett. Sharpness-aware minimization and the edge of stability, 2024.

E. Rosenfeld and A. Risteski. Outliers with opposing signals have an outsized effect on neural network optimization, 2023.

K. Wen, Z. Li, and T. Ma. Sharpness minimization algorithms do not only minimize sharpness to achieve better generalization, 2023.

M. Wortsman, G. Ilharco, S. Y. Gadre, R. Roelofs, R. Gontijo-Lopes, A. S. Morcos, H. Namkoong, A. Farhadi, Y. Carmon, S. Kornblith, and L. Schmidt. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time, 2022.

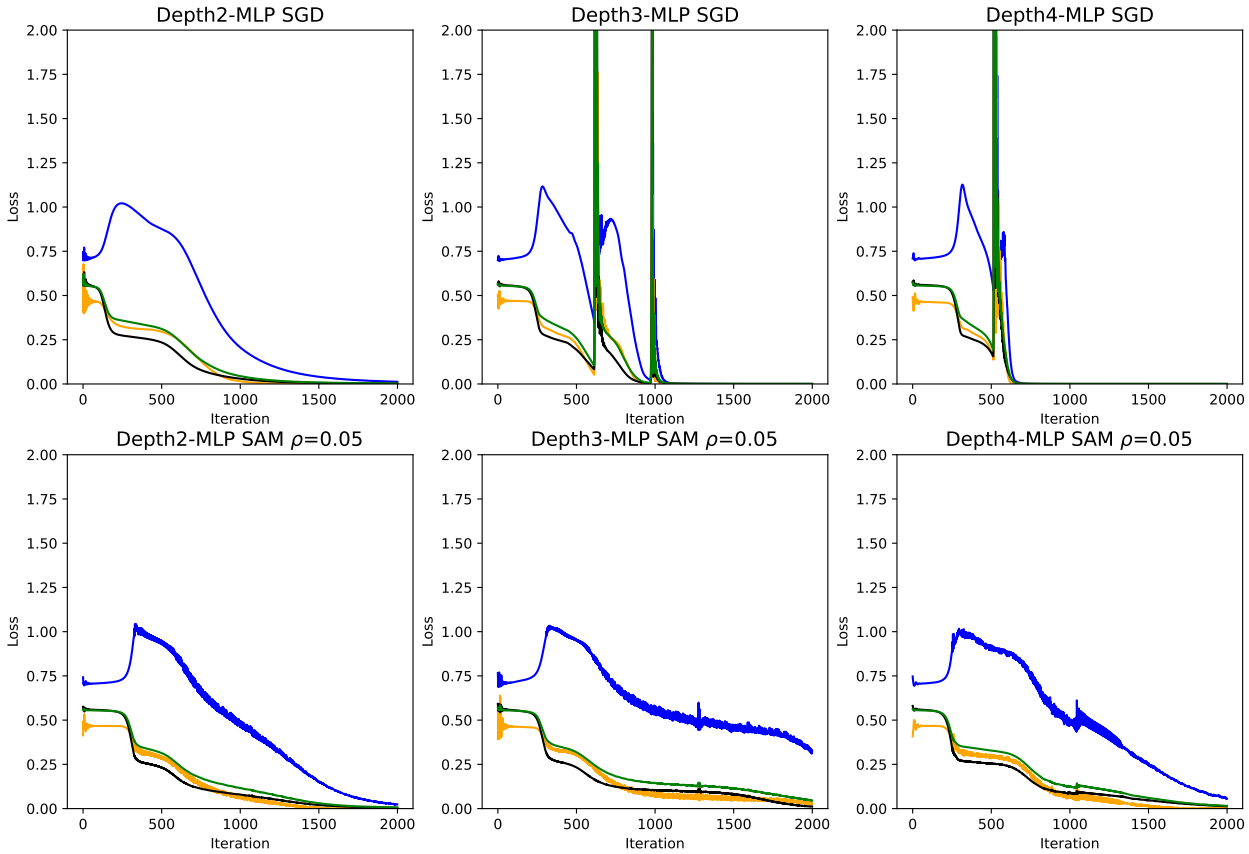


Figure 6: Loss curves for fitting $Chebyshev(5)$ using SGD and SAM.